

Pregnant Mother User Journey: Amaaii Platform.

Stage 1: Awareness & Enrollment

Step	Description
Discovery	Learns about Amaaii through CHV, clinic, social media, friend, or community outreach.
App Download/Registration	Installs Amaaii app or uses USSD/SMS-based enrollment if smartphone isn't available???? Smart Phones only for now
Consent & Orientation	Gives consent for data use and optional provider access. App explains key features in her language (Kiswahili/English). could handle this in the onboarding screen

Stage 2: Profile Setup & Risk Assessment

Step	Description	Is this being done by the Mother. do they have this data, could they need assistance from a practitioner?
Initial History Entry	Enters key health info: age, gravida, parity, LMP, chronic illnesses, past pregnancy complications. find an easy and intuitive way to fill in this data. Make it interactive	
Risk Categorization	App uses this data to assign low, medium, or high risk	
Personalized Dashboard	Her home screen shows stage-appropriate resources, journaling prompts, and upcoming milestones. MoodBoard this	

Onboarding

At enrolment, pregnant women are asked questions on socio-economic status, demographics and obstetric history including pre-existing medical conditions. Collecting detailed maternal history and personal data is **foundational** for Amaaii to truly make a difference in both **risk reduction** and **user experience personalization**. Here's how and why

1. Early Identification of High-Risk Pregnancies

Understanding a woman's:

- **Obstetric history** (e.g., past complications, stillbirths, C-sections),
- **Medical conditions** (e.g., diabetes, hypertension),
- **Age and social context** (e.g., teenage pregnancy, single motherhood),

...allows Amaaii to **flag potential risks early** in pregnancy. This enables the platform to:

- Prioritize women for urgent follow-up.
- Guide them on when and where to seek care.
- Notify caregivers or facilities if danger signs emerge.

Example: A woman with a history of pre-eclampsia can be monitored for symptoms like headaches or epigastric pain, and receive automated alerts when to seek care.

Should these symptoms be entered on the app/ detected from journaling/ both

2. Tailored Journaling & Content Experience

Every woman's pregnancy journey is unique. Amaaii can use collected data to:

- Offer **personalized journaling prompts** based on trimester, health status, and emotional needs.

- Provide **targeted health education** (e.g., gestational diabetes diet tips, pelvic floor exercises after C-section). *Could have an educational section for this as nav item*
- Suggest **mental health check-ins** and mindfulness exercises for users with anxiety or depression. *Could be on the homepage of the app/ also as part of the educational section -> give this section a suitable name (Like peace on the moodboarded inspo)*
- Customize **reminders and to-do lists** (e.g., "Pack for hospital by 36 weeks", "Ask doctor about iron levels"). *This will be in the calendar section. With one upcoming reminder on homepage*
Calendar should also include filters for symptoms, visits, immunizations etc

Example: A first-time mother in a rural setting may receive more educational prompts and encouragement around ANC visits, while a second-time mother with a history of preterm labor may receive monitoring alerts and early birth planning support.

3. Enhancing Continuity of Care

With centralized, digital documentation: *Need a dashboard for this*

- **Healthcare providers** can view a woman's evolving history for more informed clinical decisions.
- **Care companions**, doulas, or CHWs can be looped in to assist based on risk level or social vulnerability.
- Amaaii becomes a **longitudinal maternal health record** that spans preconception, pregnancy, birth, and postpartum phases.
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4. Mental Health & Psychosocial Support

Mental health screening and social history help Amaaii:

- Identify users needing emotional support.
- Offer **guided reflections**, stress management tips, and referral to tele-counseling or peer groups.
- Create a safe, affirming space through **empathetic journaling prompts**.

Need to think about how this will look like, and the kind of prompts we will use. Plus the architecture of prompt-based journaling

Example: A mother reporting low mood and weak support system may be encouraged to join a local peer support group or access postpartum depression screening tools.

Could consolidate all this data using an agentic AI to create these actionable insights and recommendations

5. Driving Behavior Change Through Nudges & Reminders

By analyzing patterns in a user's history and interaction data, Amaaii can:

- Predict care gaps (e.g., missed ANC visits).
- Send **behavioral nudges** like appointment reminders or hydration alerts.
- Reward health-promoting behaviors through **gamified encouragement** (e.g., completing journaling streaks, following self-care routines).

Streaks would be nice, plus hydration reminders

Examples of these self-care routines for pregnant mothers(Notifications)

6. Feeding into AI-Driven Public Health Insights

When anonymized and aggregated, these histories allow Amaaii to:

- Identify **community-level risk trends** (e.g., early warning of outbreaks, rising anemia cases).
- Generate insights for **policy and health system design**.
- Create a **feedback loop** that improves AI prediction and content delivery over time.

Collecting rich personal and medical information is not just a formality — it is the **engine** behind:

- Predictive maternal risk screening,
- Personalized, relevant, and compassionate care,
- Continuous engagement throughout the pregnancy journey,
- And ultimately, **reducing maternal mortality and improving maternal mental health outcomes**.

Key categories of maternal information.

1. Demographic & Social History

- **Age** (e.g. <18 or >35 = higher risk)
 - **Marital status** and **support system**
 - **Education level** (impacts health literacy)
 - **Residence** (urban/rural, county, proximity to facility) **Malaria endemic zones: based county**
 - **Health insurance** coverage???
- Current health provider/ own provider details**

2. Obstetric History

1. Gravida & Parity (G & P)

- **Gravida (G):** Total number of times pregnant (including current pregnancy)
- **Parity (P):** Number of births ≥ 20 weeks (live or stillbirth)

2. Number of Live Births

- Including full-term and preterm
- Include outcomes (e.g., vaginal or cesarean, complications, neonatal health)

3. Number of Miscarriages or Abortions

- Spontaneous (miscarriage) or induced
- Gestational age at loss

4. History of Stillbirth

- Gestational age
- Cause, if known

5. Mode of Previous Deliveries

- Vaginal
- Cesarean Section (reason for C-section)
- Assisted delivery (forceps, vacuum)

6. Pregnancy Complications in the Past

- **Pre-eclampsia / Eclampsia**
- **Gestational diabetes**
- **Preterm labor**
- **Placenta previa or abruption**
- **Postpartum hemorrhage**

7. Multiple Pregnancies

- Twins, triplets, etc.
- Outcome and complications

8. Birthweight of Previous Babies

- Large for gestational age (LGA)
- Small for gestational age (SGA)
- Low birth weight (<2.5kg)

9. Antenatal Care (ANC) in Previous Pregnancies

- Access and frequency of ANC visits
- Any barriers faced (financial, access, etc.)

10. Postpartum Complications or Experiences

- Postpartum hemorrhage, infection
- Depression, anxiety, or trauma
- Breastfeeding challenges
- Neonatal outcomes

11. Current Pregnancy Details (for continuity)

- Estimated Due Date (EDD)
- Current symptoms or concerns
- Fetal movements
- ANC attendance so far

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12. Contraceptive Use Before Current Pregnancy necessary afterwards

- Type used, duration
- Pregnancy planned or unplanned

3. Medical History

• Chronic illnesses

- Hypertension
- Diabetes
- Asthma
- Epilepsy
- Heart disease
- Sickle cell anemia
- Thyroid disorders

• HIV status

• History of surgeries (especially pelvic/abdominal)

• Mental health conditions

- Depression
- Anxiety
- Bipolar disorder

Others(any other conditions: type)

4. Medication use (including traditional/herbal remedies)

Type them up

5. Lifestyle & Behavioral Factors

- **Smoking / tobacco use**
- **Alcohol use**
- **Illicit drug use**
- **Diet and nutrition**
- **Exercise patterns**
- **Domestic violence / Intimate partner violence**

6. Accessibility & Health System Use

- **Distance to nearest health facility**
- **Means of transport during labor**
- **Birth preparedness (birth plan, packed bag, etc.)**
- **Referral history or risk flagged by clinicians**

Screens

Welcome & Consent

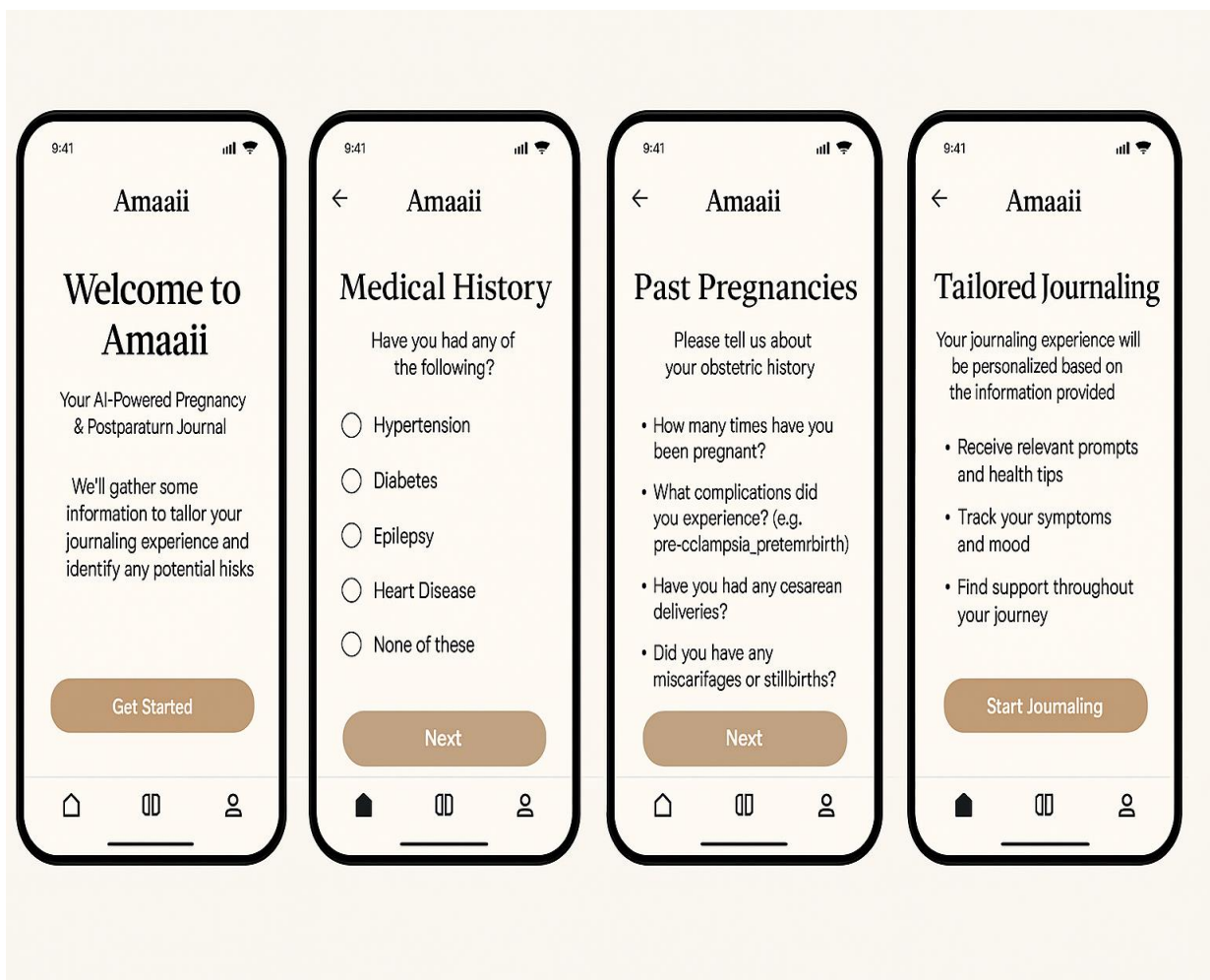
Account setup (name, contact, location, age, gestational age)

Medical History Form:

- Past pregnancies (complications, deliveries, losses)
- Pre-existing conditions (e.g., diabetes, hypertension)
- Lifestyle factors (smoking, alcohol, stress)

Risk Summary Screen

You made this?:)



Stage 3: Daily Engagement & Self-Monitoring

Google speech to text: from Joy

Step	Description
Guided Journaling	Daily/weekly prompts: “How are you feeling today?”, “Any pain?”, “What’s your mood like?”
AI Suggestions	Based on entries, Amaaii suggests tips (“rest more,” “talk to your nurse about swelling,” etc.)
Mood & Stress Check-ins	Simple emoji/mood trackers help capture mental health signals.
Wellbeing Prompts	Breathing exercises, affirmations, or nutrition tips tailored to her stage.

Journaling Interface

The core of Amaaii’s emotional and physical support system.

Options include:

Send notifications, and on opening app if not entered for the day prompt them

- Daily check-ins (How are you feeling physically/emotionally?)
- Prompts (e.g., “Did you sleep well?” “Any unusual symptoms today?”)
- Mood tracking **Moodboard**
- Free-text entries (with NLP used to detect stress, anxiety, red flags)

Why it matters: Helps flag early signs of depression, anxiety, or pregnancy-related complications.

Stage 4: Medical Milestones & Alerts

Step	Description
ANC Visit Tracker	Tracks appointments, reminds her when next one is due, and logs past visits.
Danger Sign Alerts	App prompts her to seek care if she reports red-flag symptoms like bleeding or severe headache. <i>Symptoms log should be easily accessible</i>
Baby Growth Milestones	Offers weekly fetal development updates and self-care tips. <i>On homepage, also include tips based on stage and history</i>

Stage 5: Education & Support

Nav section on this plus mental wellness

Step	Description
Culturally Relevant Content	Articles, voice notes, or animations about pregnancy, breastfeeding, birth prep, etc. in local language.
Ask-A-Midwife	Can submit a question anonymously or get connected to a provider (if available).
Peer Groups (optional)	Option to join safe, moderated group chats or forums based on trimester, region, or topic.

Chat section: groups, explore, doctor, share link

Stage 6: Provider Interaction (if opted in)

Nurse can fill in, if flagged send notifications

Step	Description
Provider Linkage	Connects her to her clinic, CHV, or nurse who can view her profile (if permission is given).
Secure Messaging	Can receive check-in messages from provider or nurse (e.g., after a flagged journal entry).
Referral Support	Receives referrals if urgent care is needed, with directions or transport support (where applicable).

Stage 7: Postpartum Transition

Maybe send a notification/something to friends/partner etc

Include this but with simplified PPD screening for the MVP first

Step	Description
Birth Journal	Captures delivery experience, birth outcome, initial baby info (e.g., weight, sex, location).
Early Motherhood Support	Switches to postpartum mode with breastfeeding tips, mental health monitoring, and danger sign education.
PPD Screening	Guided questions to help screen for postpartum depression, with referral pathways if needed.

Stage 8: Retention, Graduation & Impact

Step	Description	early days, breast feeding etc
“Graduation” Badge	Celebrates her transition out of active pregnancy care at 6+ weeks postpartum.	
Continued Learning	Option to stay for extended parenting tips and child health content.	
Feedback Loop	She can share how Amaaii helped her, suggest improvements, or invite a friend.	

End of happy path

Healthcare Provider Interface

Overview

This may end up just having the data plain and simple for the MVP stage. So no notifications for now, just simple CRUD. Focus should be on the users' app, phone and web version for MVP.

Also first just have an admin view then if time allows have the role-based access

This portal will allow approved medical professionals to **access, monitor, and support** pregnant and postpartum women enrolled in Amaaii, with **user consent** and in line with data privacy laws (e.g., Kenya's Data Protection Act).

Key Functionalities for Healthcare Provider Interface

1. Patient Dashboard

- View profiles of women under their care.
- See gestational age, risk status, vitals trends (if integrated), ANC attendance, journaling logs.
- View flagged symptoms or mental health risks based on automated assessments.

2. Medical History Access (With Consent)

- Obstetric history, chronic conditions, allergies, and current pregnancy status.

3. Real-Time Alerts

- Notifications about high-risk signs (e.g., elevated BP, bleeding, suicidal ideation).
- Suggested follow-ups or actions based on national guidelines.

4. Secure Messaging & Notes

- Chat with patients (optionally).
- Add notes visible to the patient (or internal-only).
- Ability to refer to facilities or specialists.

5. Digital ANC Toolkits

- WHO or MoH Kenya ANC visit checklist templates.
- Ability to schedule and document virtual or physical visits.
- Share educational content or videos with patients.

6. Analytics Dashboard

- Aggregate data insights: population health trends, common symptoms, follow-up adherence.
- Can be shared with hospital admins or NGOs for monitoring & evaluation.

7. Role-Based Access

- Midwives, nurses, CHVs, doctors get different access levels.
- Admin/clinical supervisors manage user roles and see audit logs.

Integration with Patient Side

- Patients can **grant/revoke access** to their provider or facility.
- Consent is recorded digitally for compliance and trust.
- Provider notes can be optionally shared in a simplified way (e.g., “Your doctor recommends an extra BP check this week”).

Privacy and Ethics

- All access must be **consent-driven** and **encrypted**.
- Offer opt-in and opt-out options for each feature (especially mental health sharing).

Healthcare Provider User Journey:

Amaaii Platform

Stage 1: Awareness & Onboarding

Need documentation on ethical standards, access, security of data, privacy

Step	Description
Discovery	Provider learns about Amaaii through a hospital program, CHV network, MoH initiative, or professional forum.
App/Web Sign-Up	Provider downloads the Amaaii Clinician App or logs into the provider portal via web.
Verification	Credentials are verified (e.g., license number, facility code, referral code) before account is approved.
Orientation	Brief onboarding guide or walkthrough introduces provider to Amaaii's core features, privacy policy, and patient consent protocols.

Stage 2: Accessing the Dashboard

Step	Description
Login & Role Access	Provider logs in and lands on the clinician dashboard, personalized by role (nurse, doctor, CHV, specialist).
Patient List	Sees a list of patients under their care or from their facility who have granted access.
Patient Search	Ability to search by ID, phone number, name, or ANC number.

Stage 3: Reviewing Patient Profiles

This should be implemented fully though, to maintain the aspect of AI-generated insights and alerts which I think is essential for the MVP, know?

Step	Description
Profile Overview	Quick view of gestational age, risk category (low/moderate/high), recent symptom entries, and mental health flags.
Medical History	Full obstetric and medical history, ANC visit summaries, medications, allergies.
Risk Flags & Alerts	AI-generated alerts for danger signs, non-compliance, depressive symptoms, etc.
Journal Insights	View patient journaling logs to assess psychosocial wellbeing or signs of distress.

Stage 4: Engagement & Intervention

Consultation fees: premium feature

Step	Description
Secure Messaging	Send and receive messages from patient with a clear audit trail (optional feature).
Notes & Recommendations	Enter clinical notes or guidance, optionally visible to patient (e.g., “increase fluids, rest more”).
Referrals	Refer patient to a higher-level facility or a specific specialist directly through the system.
Follow-Up Scheduling	Schedule next ANC visit or virtual check-in with reminders for both patient and provider.

Stage 5: Education & Empowerment

Step	Description
Share Resources	Send pre-vetted education content (e.g., videos, leaflets in Kiswahili or English) based on patient condition.
Mental Health Toolkit	Access tools to screen and guide basic mental health interventions or refer out. Premium feature: Flag and suggest visiting doctor? Forms/ doctor
Community Support Links	Suggest local support groups, nutrition clinics, breastfeeding workshops, etc. Could add a link, which is displayed in a cool way on the user's side, like on the "Mood/Peace" page have these as suggested communities and then click to open link

Stage 6: Monitoring & Reporting

Step	Description
Population Insights	View anonymized trends in their patient panel (e.g., % high-risk pregnancies, average ANC compliance).
Export Reports	For facilities working with partners or MoH: export patient logs, referrals, and outcomes. Just as an excel file for now
Feedback Loop	Provide feedback to Amaai platform to improve recommendations or request new features. Could be done on the appstore/playstore/website

Stage 7: Data Privacy & Consent Management backend

Step	Description
Consent Check	Before accessing each patient's data, verify consent was granted.
View Consent Log	See when/where patient gave or revoked consent.
Data Lock	Patient can hide certain mental health or journaling data if they don't want provider visibility (respecting autonomy).

Backend Team

Sources of data: national guidelines on this